

Researchers conducted experiments to measure the friction properties of various materials using a steel ramp with an adjustable incline angle, shown in Figure 1. They prepared blocks with different masses made of various materials. In one experiment, a block was placed on the ramp and the angle of the ramp was increased until the block just began to slide.

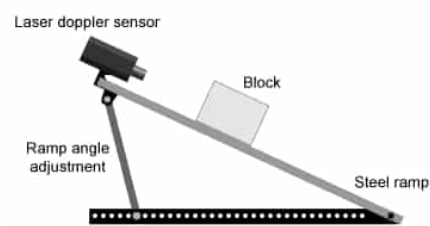


Figure 1 Schematic of experimental setup

In a second experiment, a block was positioned on the ramp and given a slight push to see if it slid down the ramp. If the block slid down the ramp, its position and speed while sliding down the ramp were monitored using a laser Doppler sensor. An example of the speed measurements from a block sliding down the ramp is shown in Figure 2.

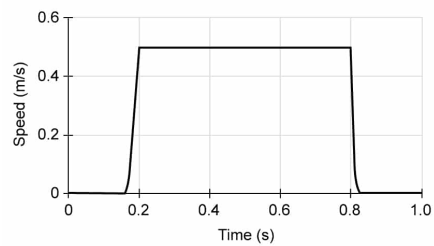


Figure 2 Speed of block sliding down ramp

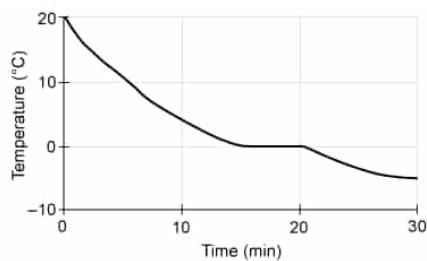
In a third experiment, blocks were pulled along a flat steel surface with a rope. The position and speed of the block were measured with the laser doppler sensor.

The researchers used the results of these experiments to calculate the coefficients of static and kinetic friction between steel and the block materials. The friction properties are summarized in Table 1.

Table 1 Coefficients of Friction for Selected Materials

Materials	Coefficients of friction	
	Static friction	Kinetic friction
Wood on steel	0.5	0.2
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Ice on steel	0.04	0.01

The researchers made blocks of ice by placing square containers of water in a freezer and monitoring the block's temperature with a thermometer, as shown in the plot below. Why was the block's temperature a constant 0 °C from 15 min to 20 min in the plot?

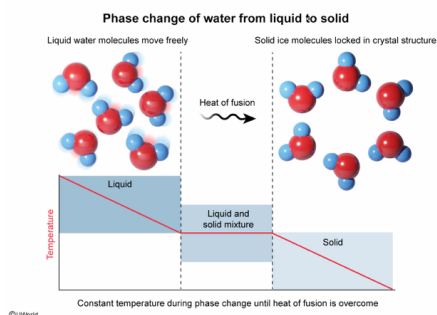


- ☐ A. The densities of water and ice are different. [3%]
☐ B. The temperature of the freezer was 0 °C. [3%]
☒ C. The water must overcome the heat of fusion. [89%]
☐ D. The water must overcome the heat of vaporization. [3%]

Correct

90% answered correctly

Explanation:



Heat represents the energy transferred between two substances with different **temperatures**. Energy is transferred from the warmer substance to the cooler substance until thermal equilibrium is reached. A substance undergoes **phase changes** when heated, passing from the solid phase to the liquid phase and finally to the gas phase.

During phase changes, energy is required to form or break bonds between molecules. Consequently, the heat energy transferred during a phase change does not alter the temperature of the material. For example, a water and ice mixture is 0 °C whether it contains 10% ice or 90% ice. The energy required to change phase between solid and liquid is the **heat of fusion**. A phase change between liquid and gas requires energy equal to the heat of vaporization.

In this question, water changes phase from liquid to solid as it is cooled. During this phase change, heat is removed from the water but the temperature remains constant. Therefore, the liquid water is transforming into solid ice during the time interval from 15 min to 20 min in the plot, and the temperature remains constant until the heat of fusion is overcome.

(Choice A) The densities of water and ice are different, but that fact does not cause the temperature to remain constant when water undergoes a change from one phase to another.

(Choice B) The temperature of the block continued to decrease to about -5 °C. Therefore, the temperature of the freezer was not 0 °C.

(Choice D) The heat of vaporization is required when changing phases between liquid and gas, not between liquid and solid.

Educational objective:

The heat of fusion is the amount of heat energy required to transform a substance between its liquid phase and solid phase. The material remains at its freezing temperature during this phase change until it is all in solid form.

Time spent: 0 sec
 Subject:
 Physics

QID: 403606
 System:
 Mechanics and Energy

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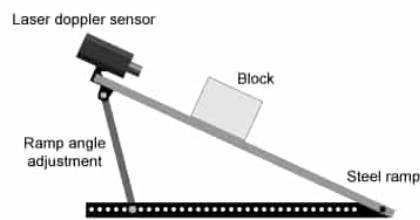


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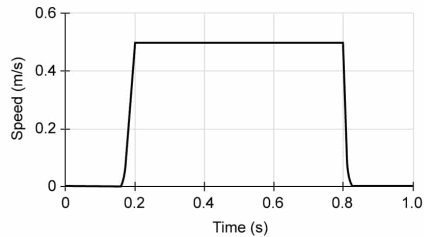


Figure 2 Speed of block sliding down ramp

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Laser light from the Doppler sensor travels through air with a refractive index of n_1 and into a glass block at an incident angle of θ_1 . The refractive index of the glass block is n_2 , and the angle of refraction is θ_2 . What is the ratio of n_1 to n_2 ?

- ☐ A. $\frac{\theta_2}{\theta_1}$ [4%]

- ✓ ☒ B. $\frac{\sin \theta_2}{\sin \theta_1}$ [73%]
- ☐ C. $\frac{\theta_1}{\theta_2}$ [4%]
- ☐ D. $\frac{\sin \theta_1}{\sin \theta_2}$ [18%]

Correct

74% answered correctly

Explanation:

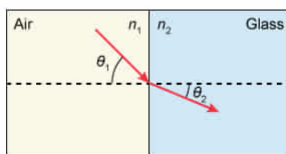
Light refraction: Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\frac{n_1}{n_2} = \frac{\sin \theta_2}{\sin \theta_1}$$

- n_1 Refractive index of air
- θ_1 Incident angle of light
- n_2 Refractive index of glass
- θ_2 Refractive angle of light

Laser light traveling from air to glass



©UWorld

Light traveling between two materials with different **refractive indexes**, n_1 and n_2 , undergoes **refraction** as it passes from the first material into the second, bending the light ray when it crosses the boundary. The refractive index depends on the **speed of light** in the material, with a greater refractive index indicating a slower speed of light in that material. The incident angle θ_1 and refracted angle θ_2 , which are both measured relative to the normal of the boundary (a line perpendicular to the boundary), are given by **Snell's law**:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

In this question, the laser light is refracted as it passes from air into the glass block. The ratios of the refractive indexes of the two materials are calculated by rearranging Snell's law:

$$n_1 = \frac{n_2 \sin \theta_2}{\sin \theta_1}$$

$$\frac{n_1}{n_2} = \frac{\sin \theta_2}{\sin \theta_1}$$

Therefore, the ratio of n_1 to n_2 equals $\frac{\sin \theta_2}{\sin \theta_1}$.

(Choices A and C) The ratio of the refractive indexes equals the ratio of the sines of the angles, not the ratio of the angles themselves.

(Choice D) $\frac{\sin \theta_1}{\sin \theta_2}$ equals the ratio of n_2 to n_1 , not the ratio of n_1 to n_2 .

Educational objective:

The refraction of light is given by Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$. This equation can be rearranged to show that the ratio of n_1 to n_2 equals $\frac{\sin \theta_2}{\sin \theta_1}$.

Time spent: 0 sec
Subject:
Physics

QID: 403607
System:
Mechanics and Energy

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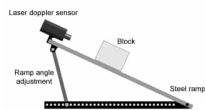


Figure 1 Schematic of experimental setup

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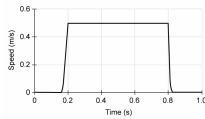


Figure 2 Speed of block sliding down ramp

In a third experiment, blocks were pulled along a flat steel surface with a rope. The position and speed of the block were measured with the laser doppler sensor.

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Table 1 Coefficients of Friction for Selected Materials

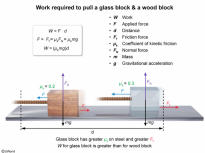
Materials	Coefficients of friction	
	Static friction	Kinetic friction
Wood on steel	0.5	0.2
Steel on steel	0.8	0.4
Glass on steel	0.7	0.3
Ice on steel	0.04	0.01

A block made of glass and a block made of wood have the same mass. Each block is pulled by a rope at constant speed the same distance along a horizontal steel surface. Would the same amount of work be required to pull the glass block as the wood block?

- ☐ A. Yes, because the masses of the blocks are the same [8%]
☐ B. Yes, because the blocks are pulled along a horizontal surface [1%]
☐ C. No, because the coefficients of static friction for the block materials on steel are different [18%]
☒ D. No, because the coefficients of kinetic friction for the block materials on steel are different [71%]

Correct
71% answered correctly

Explanation:



The **work** W required to pull an object with a constant force F along a horizontal surface equals the product of F and the distance d over which the object is pulled:

$$W = F \cdot d$$

In this question, the blocks are pulled with **constant speed** along a steel surface. **Newton's first law of motion** implies that the net force F_{net} on an object is zero when it moves at constant speed. Hence, the **kinetic friction force** F_f must be equal but opposite to F :

$$F_{\text{net}} = F - F_f = 0$$
$$F = F_f$$

Furthermore, F_f is the product of the **coefficient of kinetic friction** μ_k and the **normal force** F_N exerted by the surface:

$$F_f = \mu_k F_N$$

For a horizontal surface, F_N equals the object's **weight** (ie, the product of the object's mass m and gravitational acceleration g):

$$F_N = mg$$

Consequently, F_f equals:

$$F_f = \mu_k mg$$

Thus, W equals:

$$W = (\mu_k mg)d$$

For both blocks, the values for m , g , and d are the same. However, Table 1 states that μ_k for glass on steel is 0.3 but μ_k for wood on steel is 0.2. Therefore, the W required to slide the glass block is greater because μ_k for glass on steel is 50% greater than for wood on steel.

(Choices A and B) The amount of work required to pull the blocks is not the same. Although the blocks have the same mass and are pulled along the same horizontal surface, pulling the glass block requires more work because its coefficient of kinetic friction on steel is greater.

(Choice C) The coefficients of static friction are not relevant because the blocks are sliding along a surface.

Educational objective:

The work required to pull an object at constant speed along a surface equals the product of the coefficient of kinetic friction, the object's weight, and the distance over which the object is pulled.

Time spent:0 sec
Subject:
Physics

QID:403608
System:
Mechanics and Energy

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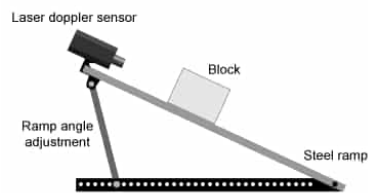


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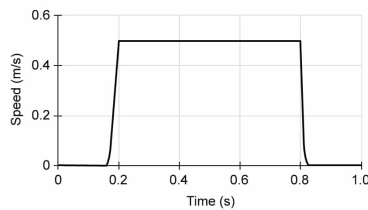


Figure 2 Speed of block sliding down ramp

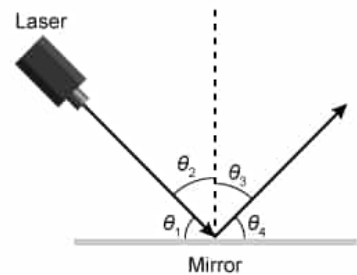
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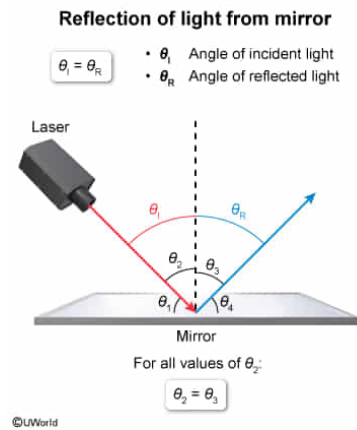
A mirror is placed at the back of the block during the experiment. The laser light from the Doppler sensor reflects off the mirror, as shown in the diagram below. (Note: The figure is NOT drawn to scale.) Which of the following equations is true for any value of the incident angle, θ_2 ?



- ☐ A. $\theta_1 = \theta_2$ [1%]
- ☐ B. $\theta_1 = \theta_3$ [14%]
- ☒ C. $\theta_2 = \theta_3$ [83%]
- ☐ D. $\theta_3 = \theta_4$

Correct
83% answered correctly

Explanation:



All surfaces **reflect** light. Rough surfaces, such as a person or a tree, reflect light in all directions. A very smooth surface, such as a **mirror**, only reflects light in one specific direction. For a mirror, the **reflected angle** θ_r of the light is equal to the **incident angle** θ_i of the light, with the angles measured relative to a line perpendicular to the surface of the mirror (ie, the normal to the surface):

$$\theta_i = \theta_r$$

In this question, θ_i of the light is θ_2 and θ_r is θ_3 . Therefore, θ_2 is always equal to θ_3 .

(Choice A) θ_1 is not always equal to θ_2 . The sum of θ_1 and θ_2 must equal 90° because the dotted line in the figure forms a right angle with the mirror's surface, but that does not mean the angles are equal for all values of θ_2 .

(Choice B) θ_1 is not always equal to θ_3 . Because θ_3 is equal to θ_2 , the sum of θ_1 and θ_3 must equal 90° , but that does not mean θ_1 equals θ_3 for all values of θ_2 .

(Choice D) θ_3 is not always equal to θ_4 . The dotted line in the figure and the mirror's surface form a right angle, so the sum of these angles must be 90° , but they are not equal for all values of θ_2 .

Educational objective:

Reflection occurs when light encounters an interface between two different media and the light travels within the original medium but in a new direction. For smooth interfaces, the angle of reflection equals the angle of incidence.

Time spent: 0 sec
Subject:
Physics

QID: 403609
System:
Mechanics and Energy

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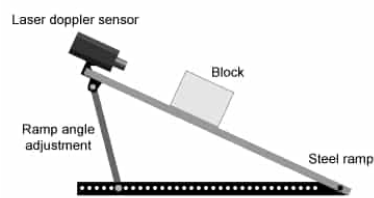


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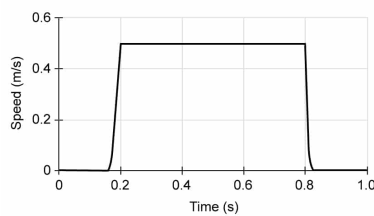


Figure 2 Speed of block sliding down ramp

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Ice on steel	0.04	0.01

A 5 kg block of iron slides down a 30° steel ramp at a constant velocity. What is the approximate value for the coefficient of kinetic friction for iron on steel? (Note: $\sin 30^\circ = 0.50$, $\sin 60^\circ = 0.87$, $\cos 30^\circ = 0.87$, and $\cos 60^\circ = 0.50$.)

- ☐ A. 0.1 [8%]
- ☒ B. 0.6 [63%]
- ☐ C. 0.8 [18%]
- ☐ D. 1.0 [9%]

Correct
63% answered correctly

Explanation:

Coefficient of kinetic friction: block sliding on an incline

$$\sum F_x = 0 = F_{wx} \sin \theta - F_k$$
$$F_k = \mu_k F_n$$
$$\sum F_x = 0 = F_{wx} \sin \theta - \mu_k F_n \cos \theta$$

- F_x Forces parallel to incline
- F_w Weight of block
- θ Angle of incline
- F_k Kinetic friction force on block
- μ_k Coefficient of kinetic friction
- F_n Normal force
- F_y Forces perpendicular to incline

Iron block
Steel ramp
 $\theta = 30^\circ$
Constant velocity

$$F_{wx} \sin \theta = \mu_k F_n \cos \theta$$
$$F_{wx} \sin \theta = \mu_k F_{wx} \cos \theta$$
$$\mu_k = \frac{F_{wx} \sin \theta}{F_{wx} \cos \theta} = \frac{\sin \theta}{\cos \theta}$$
$$\mu_k = \frac{\sin 30^\circ}{\cos 30^\circ} = \frac{0.50}{0.87} \approx 0.6$$

An object moving at constant velocity is in **dynamic equilibrium**, and all the forces on the object are balanced. In this question, the iron block moves at constant velocity down the incline. Hence, the sum of the parallel forces $F_{\parallel}$ and the sum of the perpendicular forces $F_{\perp}$ relative to the incline must equal zero.

$F_{\parallel}$ consists of the force F_k of **kinetic friction** between the block and the ramp and the product of the **weight** F_W of the block and the sine of the angle θ of the incline. Therefore, the sum of $F_{\parallel}$ is:

$$\sum F_{\parallel} = F_W \sin \theta - F_k = 0$$

Furthermore, F_k equals the product of the **coefficient of kinetic friction** μ_k and the **normal force** F_N the ramp exerts on the block:

$$F_k = \mu_k F_N$$

Substituting the equation above into the first equation and solving for μ_k yields:

$$\begin{aligned} F_W \sin \theta &= F_k \\ F_W \sin \theta &= \mu_k F_N \\ \mu_k &= \frac{F_W \sin \theta}{F_N} \end{aligned}$$

The forces contributing to $F_{\perp}$ are F_N and the product of F_W and the cosine of θ :

$$\sum F_{\perp} = F_N - F_W \cos \theta = 0$$

Rearranging this equation to solve for F_N gives:

$$F_N = F_W \cos \theta$$

Substituting this equation into the above equation for μ_k yields:

$$\mu_k = \frac{F_W \sin \theta}{F_W \cos \theta} = \frac{\sin \theta}{\cos \theta}$$

Therefore, because θ equals 30° , μ_k is equal to:

$$\mu_k = \frac{\sin 30^\circ}{\cos 30^\circ} = \frac{0.50}{0.87} \approx 0.6$$

(Choice A) 0.1 is obtained by calculating the ratio of the sine of 30° and the normal force. However, the coefficient of kinetic friction is the ratio of the friction force and the normal force.

(Choice C) 0.8 is obtained by assuming the coefficient of kinetic friction equals the coefficient for steel on steel given in Table 1. However, this block is made of iron, not steel.

(Choice D) 1 is obtained by calculating the ratio of the weight of the block and the normal force. However, the coefficient of kinetic friction is the ratio of the friction force and the normal force.

Educational objective:

A block sliding down a ramp at a constant velocity is in dynamic equilibrium, and all the forces on the block are balanced. The coefficient of kinetic friction between the block and the ramp depends only on the angle of the incline.

Time spent: 0 sec
Subject:
Physics

QID: 403610
System:
Mechanics and Energy

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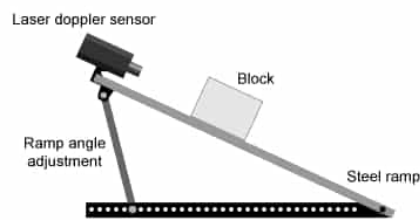


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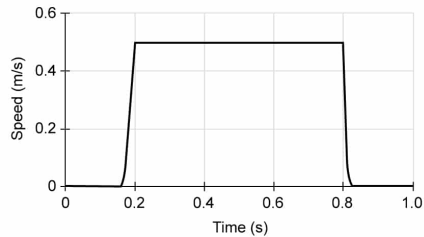


Figure 2 Speed of block sliding down ramp

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The laser Doppler sensor is positioned at the top of the ramp and emits light at 500 nm toward a block sliding down the ramp. The sensor records the data shown in Figure 2. During the interval from 0.3 s to 0.7 s, the wavelength of the light reflected back to the sensor is:

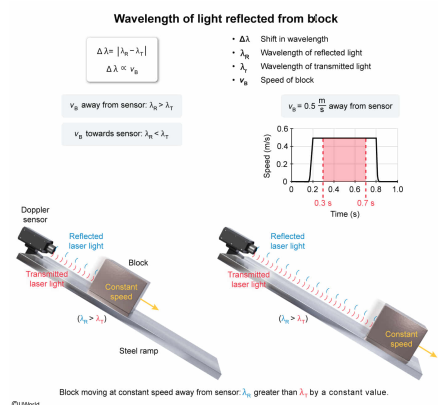
- ✔ ☒ A. greater than the emitted light by a constant value. [35%]

- ☐ B. less than the emitted light by a constant value. [32%]
- ☐ C. continuously increasing with time. [20%]
- ☐ D. continuously decreasing with time. [11%]

Correct

36% answered correctly

Explanation:



The **Doppler effect** occurs when a periodic signal emitted from a source (eg, **sound**, **light**) is perceived by an observer to have a different **wavelength** λ due to the relative motion between the source and the observer. The magnitude of this change in λ is proportional to the relative speed v between the source and observer:

$$\Delta \lambda \propto v$$

If the source is moving **toward the observer**, the perceived λ of the signal **decreases** because the source is closer to the observer for each subsequent peak in the wave, and thus the waveform is compressed. If the source is moving **away from the observer**, λ **increases** because the source is farther away for each subsequent peak in the wave, and

thus the waveform is stretched.

In this question, the laser Doppler sensor transmits light toward the block and records the reflected signal to determine its speed. Because the reflected signal is recorded by the sensor, the block acts as the source and the sensor acts as the observer.

The speed of the block in Figure 2 during the interval from 0.3 s to 0.7 s is a *constant* 0.5 m/s. Therefore, the Doppler shift in the observed λ will be constant over time. Moreover, because the block is *moving away from the sensor*, the λ of the light recorded by the sensor will be greater than the λ of the light originally emitted by the sensor.

(Choice B) The reflected wavelength will be greater than the original wavelength because the block is moving away from the sensor.

(Choices C and D) The reflected wavelength will be constant because the speed of the block does not change during the interval of interest.

Educational objective:

The Doppler effect occurs when the observed reflected wavelength of light is shifted compared to the original wavelength due to the relative motion of the source and observer. The shift in wavelength is proportional to the relative speed, and the reflected wavelength increases when the source and observer move away from each other.

Time spent: 0 sec

Subject:
Physics

QID: 403611

System:
Mechanics and Energy